

MEMNAR WEB SERVER

Web Server for Mutual Exclusivity Analysis in Cancer Genomics



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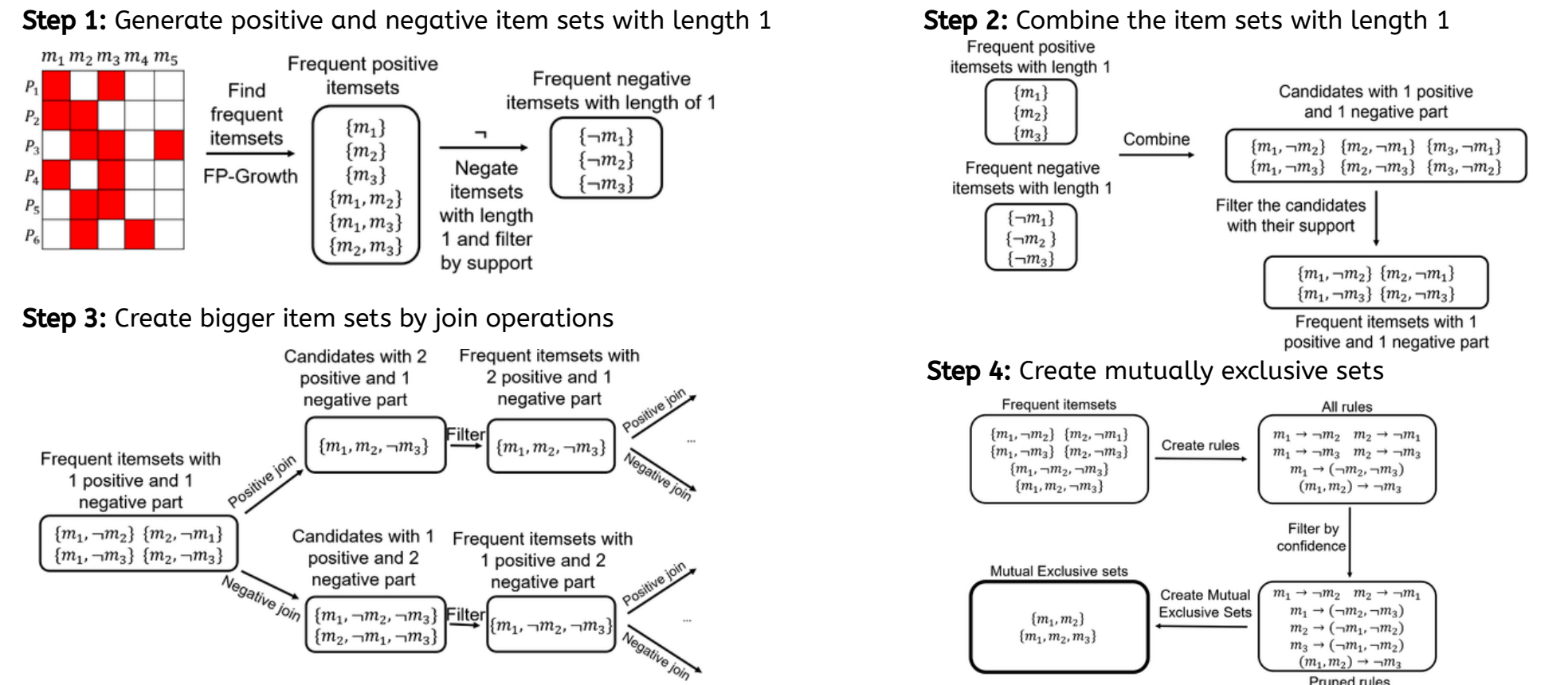
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Introduction



MEMNAR is a bioinformatics algorithm designed to discover mutually exclusive and conditional mutually exclusive mutation patterns in cancer genomics data through negative association rule mining.^[1] MEMNAR identifies relationships between gene mutations that rarely occur together within the same tumor sample by analyzing somatic mutation datasets. These mutation patterns are important for understanding cancer development, identifying driver pathways and revealing biologically significant interactions between genes. MEMNAR provides efficient analysis and supports the discovery of complex mutation relationships across different cancer datasets compared to existing approaches. The official paper is available by the QR code above.

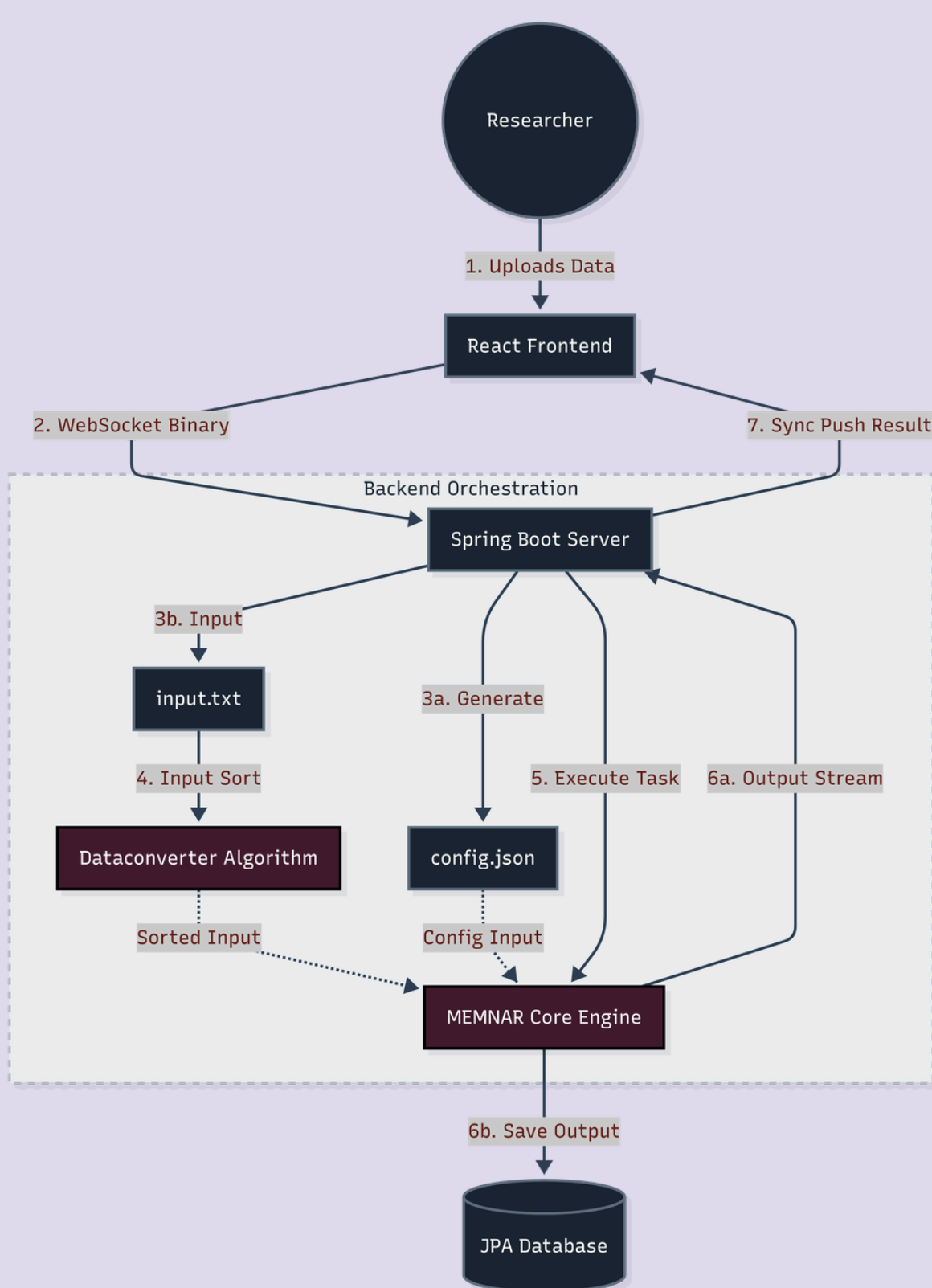


The Main Problem and The Purpose

The MEMNAR algorithm operates through command line execution and configuration files, which limits accessibility for researchers without technical expertise with CLI. The objective of this project is to develop an interactive web based platform for MEMNAR that simplifies dataset upload, parameter configuration, algorithm execution and result visualization. This platform aims to make mutual exclusivity analysis more accessible, user-friendly and manageable through a modern web interface while supporting multiple datasets and interactive analysis workflows.

Methodology

- MEMNAR algorithm has been moved to Maven dependency management system for easier integration.^[2] This has also provided the needed infrastructure for future steps.
- A Spring Boot backend and React frontend service has been developed for the management of the MEMNAR algorithm via a responsive UI.
- An interactive frontend interface was designed to allow dataset upload, parameter configuration and execution of MEMNAR through a user-friendly web platform.



- Intermediate logging monitoring system has been developed for observation of the algorithm during execution. Also for a more inclusive environment, input uploading system has been developed with a dependent algorithm for MEMNAR, DataConverter. This way the researchers will not be required to filter and sort their data manually before the algorithm execution.
- System architecture was improved to support multiple concurrent users, isolated execution workflows and scalable web-based cancer genomics analysis.

User Interface and Results

The MEMNAR Web Server is designed for easy-to-use genomic analysis, specifically targeting the needs of cancer researchers who may lack the technical expertise to use Command Line Interfaces (CLI). By abstracting the technical complexities into an intuitive web interface, the cognitive and operational overhead typically associated with bioinformatics tools are significantly reduced. Allowing for domain experts to redirect their focus from system administration toward their research.

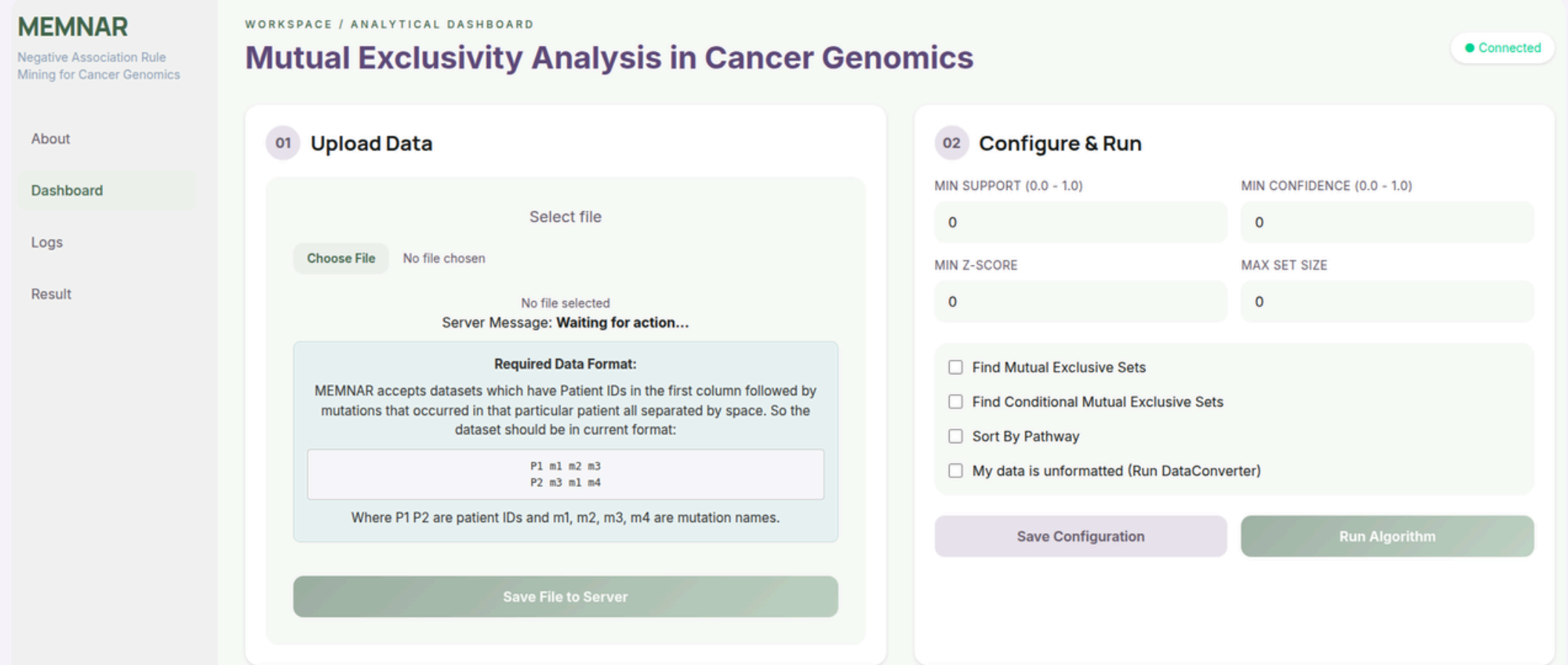
03 Process Logs

```
[12/05/2026 14:21:03] MEMNAR process started
[12/05/2026 14:21:04] Loading configuration
[12/05/2026 14:21:05] Dataset loaded: mutation_data
[12/05/2026 14:21:06] Patients: 100
[12/05/2026 14:21:06] Mutations: 34
[12/05/2026 14:21:07] Filtering started
[12/05/2026 14:21:08] 542 candidate sets generated
[12/05/2026 14:21:09] 61 sets remaining after support filtering
[12/05/2026 14:21:10] 18 sets with 1 positive 1 negative
[12/05/2026 14:21:10] 7 sets with 1 positive 2 negative
[12/05/2026 14:21:11] 3 sets with 2 positive 1 negative
[12/05/2026 14:21:12] Rule generation started
[12/05/2026 14:21:13] 41 rules generated
[12/05/2026 14:21:14] 9 rules remaining after confidence filtering
[12/05/2026 14:21:15] Mutual exclusive set generation started
[12/05/2026 14:21:16] 3 mutual exclusive candidate sets found
[12/05/2026 14:21:16] FINISHED | elapsed=13184ms
```

Process logs display intermediate stages of MEMNAR pipeline during execution. The system reports dataset loading, candidate set generation, filtering steps, rule generation and mutually exclusive set detection. Allowing users to monitor the algorithm execution.

System Architecture and Workflow

For two-way communication between client and server, MEMNAR Web Server utilizes a WebSocket-based event-driven architecture rather than traditional RESTful polling.^[3] By minimizing 'chatter', redundant header transmissions and unnecessary server pings are eliminated. Using the persistent, stateful connection, the server autonomously pushes results to the client upon algorithm completion and other significant milestones, This ensures smooth user experience and execution transparency. User configurations and analysis history are managed via Spring Data JPA.^[4]



Conclusion

The MEMNAR Web Server democratizes complex genomic analysis by removing the need for Java expertise and manual environment setup. By wrapping algorithms that require technical expertise in a web-native interface, operational overhead is minimized, allowing researchers to focus entirely on their research. With WebSockets and Spring Data JPA, the system delivers an efficient and asynchronous workflow. Ultimately, how modern web orchestration can transform sophisticated bioinformatics tools into accessible and scalable platforms for cancer research is demonstrated.

References

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